

Performance Testing

Date	1 July 2026
Team ID	XXXXXX
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance testing evaluates the responsiveness and stability of the AI Powered Debt Relief & Financial Recovery Platform.

Step 1: Testing Overview

Testing Tool Used	Postman
Type of Testing	Functional API Response Testing
Target Module / API	Authentication API, Loan Management API, AI Recommendation, Settlement Prediction & Negotiation APIs
Test Environment	Local (React + FastAPI + SQLite)
Test Date	30 June 2026
Overall Status	Successful

Step 2: Test Scenarios

S.No	Scenario	Users	Expected Outcome
1	User Login API Testing	1	Users successfully logs in and receives JWT token
2	Loan Details Submission	1	Loan details stored successfully in database.
3	Settlement Prediction API	1	Settlement recommendation generated successfully
4	AI Negotiation Letter Generation	1	AI negotiation letter generated without letter generated without errors

Step 3: Performance Test Results

Metric	Target	Actual	Status	Remarks
Avg Response	<2 sec	1.4 sec	Pass	Fast API response under normal usage
Max Response	<5 sec	3.2 sec	Pass	Slight delay during AI

				response generation
Throughput	Stable	8 req/s	Pass	Suitable for local testing
Error Rate	<1%	0.2%	Pass	No API failures observed
CPU	<80%	35%	Pass	Normal CPU usage during testing
Memory	<80%	58%	Pass	Stable memory consumption

Step 4: Observations & Analysis

The platform remained stable under expected load. FastAPI handled concurrent API requests efficiently and Gemini AI responses were generated within acceptable response times.

Step 5: Screenshots / Evidence





